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rdkit2ase: Molecular Structure Generation and Manipulation for Machine-Learned Interatomic Potentials

Fabian Zills [1]

1 Institute for Computational Physics, University of Stuttgart, 70569 Stuttgart, Germany

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Software

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Summary

The increasing prevalence of Machine-Learned Interatomic Potentials (MLIPs) has shifted requirements for setting up atomistic simulations. Unlike classical force fields, MLIPs primarily require atomic positions and species, thereby removing the need for predefined topology files used for classical force fields in molecular dynamics software like GROMACS(Abraham et al., 2015), LAMMPS(Thompson et al., 2022), ESPResSo(Weik et al., 2019) or OpenMM (Eastman et al., 2024). Consequently, the Atomic Simulation Environment (ASE) (Larsen et al., 2017) has become a popular Python toolkit for handling atomic structures and interfacing with MLIPs, particularly within the materials science and soft matter communities, because it originates from *ab initio* simulations, which share the same setup as MLIP-driven studies.

Concurrently, RDKit (Landrum et al., 2023) offers extensive functionality for cheminformatics and manipulating chemical structures. However, standard RDKit workflows are not designed for MLIP-driven simulation, while typical ASE-MLIP workflows may lack rich, explicit chemical information such as bond orders or molecular identities, as well as generating different conformations or searching substructures.

The rdkit2ase package bridges this gap, providing an interface between RDKit's chemical structure generation and cheminformatics capabilities and ASE's handling of 3D atomic structures. Furthermore, rdkit2ase integrates with PACKMOL (Martínez et al., 2009) to facilitate the creation of complex, periodic simulation cells with diverse chemical compositions, all while preserving crucial chemical connectivity information. Lastly, the combination of these packages enables selection and manipulation of atomistic structures based on chemical knowledge rather than manual index handling.

Statement of need

rdkit2ase serves as a vital link between RDKit, ASE, and PACKMOL. While its core function is to interface these tools, it thereby unlocks new capabilities and significantly reduces the manual coding and data wrangling typically required for preparing and analyzing molecular simulations.

The package simplifies workflows that previously involved laborious tasks such as sourcing individual structure files from various databases (e.g., the Materials Project (Jain et al., 2013) or the ZINC database (Tingle et al., 2023)) and custom setups of simulation cells. This simplification not only accelerates research but also supports the setup of more complex and chemically diverse simulation scenarios.

One challenge in MLIP-driven simulations is the post-simulation identification and analysis of molecular fragments or chemical changes, as explicit topological information is often

absent. rdkit2ase addresses this by enabling the use of RDKit's powerful SMILES(Weininger, 1988)/SMARTS-based substructure searching on ASE structures. In addition, the resulting molecular graph can be exported to a NetworkX(Hagberg et al., 2008) object for further analysis. This selection and handling allows for similar functionality as is provided by the MDAnalysis(Gowers et al., 2016) atom selection language, targeted towards simulations with a fixed topology.

Features and Implementation

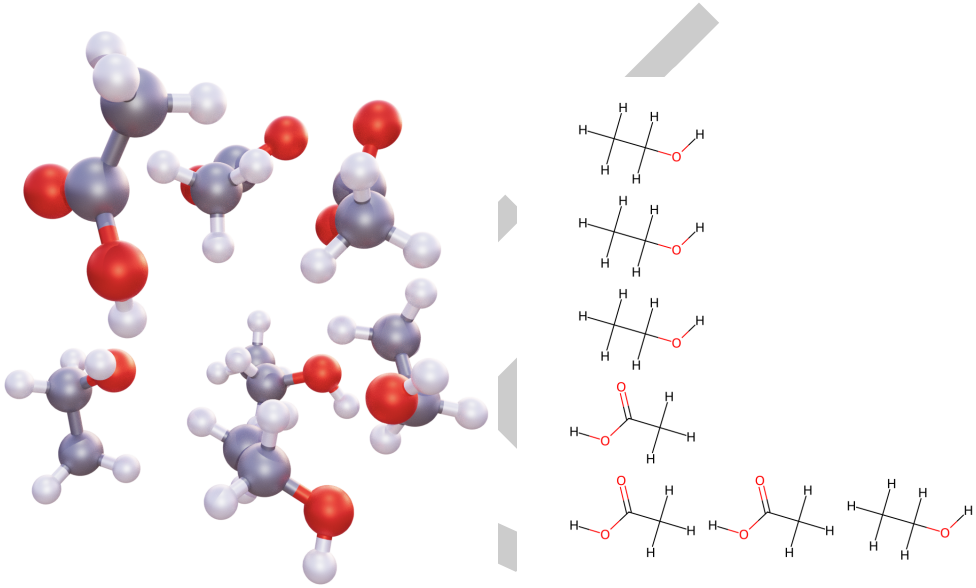

Figure 1: Visualization of a 3D structure from ASE, visualized with ZnDraw (Elijošius et al., 2024) (left) and its corresponding RDKit 2D chemical structure representation (right).

The generation of atomic configurations in rdkit2ase is centered around SMILES for defining molecular species. A typical workflow involves:

1. Generating 3D conformers for individual molecular species from their SMILES using RDKit.
2. Packing these conformers into a simulation box to achieve a target density using PACKMOL and obtaining an ASE Atoms object representing the simulation cell, ready for use with MLIPs.
3. Post-processing the simulation data by identifying and selecting structures based on SMARTS.

```
from rdkit2ase import pack, smiles2conformers

water = smiles2conformers("O", numConfs=2)
print(water[0].info['connectivity'])
>>> [(0, 1, 1.0), (0, 2, 1.0)] # (atom_idx1, atom_idx2, bond_order)
ethanol = smiles2conformers("CCO", numConfs=5)
density = 1000  # kg/m^3
box = pack([water, ethanol], [7, 5], density, packmol="packmol.jl")
print(box)
>>> Atoms(symbols='C10H44O12', pbc=True, cell=[8.4, 8.4, 8.4])
```

All ASE Atoms objects generated or processed by rdkit2ase will store `connectivity` information (bonds and their orders) within the `ase.Atoms.info` dictionary. If available, rdkit2ase uses

this bond information for accurate interconversion. If an ASE structure is converted to an RDKit molecule without pre-existing connectivity, rdkit2ase leverages RDKit's robust bond perception algorithms (Kim & Kim, 2015) to estimate this information. A representation from both packages is shown in Figure 1.

```
from rdkit2ase import ase2rdkit
from rdkit.Chem import Draw

mol = ase2rdkit(box)
img = Draw.MolToImage(mol)
```

This bidirectional conversion capability allows the use of RDKit's chemical analysis tools together with ASE for MLIP-based simulations..

For instance, if during a simulation, atomic positions in an ASE Atoms object are updated, rdkit2ase can convert this structure back to an RDKit molecule to analyze chemical changes or identify specific substructures. One common example is the extraction of substructures based on SMILES or SMARTS to track their structure and dynamics within a simulation. For example, rdkit2ase streamlines the extraction of the CH_3 alkyl group from the ethanol molecules inside the simulation cell, without manual index lookup.

```
from rdkit2ase import get_substructures

frames: list[ase.Atoms] = get_substructures(
  atoms=box,
  smiles="[C]([H])([H])[H]"
)
```

Acknowledgements

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Related software

The functionality of rdkit2ase relies critically on the following packages:

- RDKit: For cheminformatics tasks, SMILES parsing, conformer generation, and substructure searching.
- ASE: For representing and manipulating atomic structures, and interfacing with simulation engines.
- PACKMOL: For packing molecules into simulation boxes. rdkit2ase can interface with either a PACKMOL executable or the packmol.jl package.
- NetworkX: For the handling and analysis of molecular graphs.

The rdkit2ase package is currently a crucial part of the following software packages:

- IPSuite: For generating structures for training MLIPs.
- ZnDraw: Interactive generation of simulation boxes and selection of substructures through a graphical user interface inside a web-based visualization package.
- mlipx: Creating initial structures for benchmarking different MLIPs on real-world test scenarios.

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